

Underwater Basket Weaving: The Game - Game Design Document

Ethan Lindsay

Concept

The player controls a small deep sea diver gathering materials from the ocean floor before his breath runs out, then weaves the gathered materials into a basket using a timed directional input sequence. The final score reflects both how much was gathered and how precisely the basket was woven, so neither phase can be skipped or ignored.

Genre

A two-phase arcade hybrid: a real-time collection phase paired with a directional pattern-matching, Simon Says/QTE-style phase, tied together by a 3-star scoring system.

What You Are Learning / What the Game Teaches

- Resource management under a hard, real-time constraint.
- Risk/reward judgment: whether to keep exploring for more materials or head to the exit with what's already gathered.
- Fast directional reaction and pattern matching, escalating from a flat, generous timing window to a shrinking one mid-sequence.

Rules

- The breath timer counts down in real time from scene start.
- Walking into a material removes it and increments the collected count; nothing is required to "use" a collected material beyond this.
- The collection phase ends when breath reaches zero OR the player reaches the exit point, whichever comes first.
- The weave phase presents a randomized sequence of Left/Right/Up/Down prompts, one at a time, each with a time window to press the matching key.
- A correct press, a wrong press, or a timeout with no press all advance to the next prompt; only a correct press counts as a hit.
- Final score = weave accuracy $\times$ (materials collected $\div$ total materials available). This formula means a perfect weave with zero materials collected scores 0, and a full collection with a failed weave also scores near 0 — neither phase alone can carry the score.

- Stars are assigned from the final score: 3 stars at ≥ 0.75 , 2 stars at ≥ 0.5 , 1 stars at ≥ 0.25 , 0 stars below that.
- Level 0 (tutorial): open arena, 5 materials, a short 10 prompt weave sequence with a single generous flat timing window, and on-screen instructions for both phases.
- Level 1: open arena, 20 materials, a flat-timed 20-prompt weave sequence.
- Level 2: a maze-layout arena (same collection rules, navigated rather than open), 25 materials, and a 35-prompt weave sequence where the per-prompt time window drops from 1.2s to 0.7s at the halfway point, telegraphed by a red visual cue and on-screen warning.

Core Mechanics

- **Top-down movement:** a `CharacterBody2D` driven by `Input.get_vector`, standard stuff
- **Breath-depletion timer:** the collection phase's central pressure system; which is meant to prevent a "do nothing" strategy. Standing still drains breath and if the player grabs no materials, the scoring formula will convert that directly into a zero score.
- **Area2D material pickups:** each collectible is an independent scene instance with its own collision and sprite crop, removed and counted on contact.
- **Directional weave minigame:** a randomized prompt sequence with a per-prompt timing window. Wrong or missed inputs are explicit misses, not neutral, which prevents a button mashing strategy from trivializing the phase.
- **Multiplicative scoring:** weave accuracy $\times$ materials ratio. This is a deliberate design answer to "what forces participation". Ignoring either phase caps the score near zero, so passive play in either phase cannot produce a good outcome.
- **Per-level escalation:** Level 2 introduces maze navigation and an accelerating, telegraphed timing window as a harder application of the same mechanics taught in the tutorial and Level 1.
- **Basket result feedback:** four distinct basket sprites (great/good/bad/fail) tied directly to the star outcome, giving the player a visual, thematic result rather than only a number.

Genre & Background Information

The original idea for this game came after I decided a 2D platformer was too simple and hard to make interesting. So, the age old joke of underwater basket weaving came to me and I decided to try and run with it. I decided to still stick with more of an arcade style game and it kind of evolved into a puzzle game through ideation and me trying to think of ways to add difficulty to the core gameplay loop.

Design Schemas (Rules of Play)

Rules schema (the game as a formal system): the breath timer, collection count, weave sequence, and the multiplicative scoring formula are all closed, well-defined systems. Every outcome is a function of player input and time, with explicit thresholds (0.25/0.5/0.75) mapping score to stars.

Play schema: the tension between exploring for more materials versus retreating to weave is a moment-to-moment decision, not a rule explicitly stated to the player. It emerges from feeling the breath timer and seeing remaining materials. The weave phase's crunch-time visual cue adds a sense of urgency on top of a rules-level change when the timing window is shortened.

Culture schema: A game about underwater basket weaving in itself is more of a joke concept but it was genuinely fun working on the game and appreciating the craftsmanship of basket sprites.

Description of Changes Between Iterations

- **Genre pivot:** this project began as a 2D platformer (starfish collecting parts to build a spaceship, it was gonna be awesome) and was pivoted to the current collection-and-weave concept partway through development, after concluding the platformer leaned too heavily on pre-existing genre conventions, like Mario and Kirby, to differentiate cleanly within the project timeline.
- **Scoring formula iteration:** considered a hard minimum-materials threshold and a materials-determine-sequence-length model before settling on the multiplicative $\text{weave-accuracy} \times \text{materials-ratio}$ formula, chosen specifically because it structurally prevents passive play in either phase without needing an arbitrary cutoff. I was really trying to think of ways to keep Dr. Walter from breaking my game.
- **Tileset and collision debugging:** early iteration involved correcting floor tiles that were unintentionally solid, fixing shared-resource bugs across duplicated material instances, and separating per-level scripts after an initial version let edits to one level's logic unintentionally affect another. Most debugging and playtesting was done as development was ongoing.
- **Level 2 escalation design:** iterated on the weave phase's difficulty curve through direct playtesting rather than fixed upfront values, arriving at a 32-prompt sequence with a 1.2s→0.7s timing drop at the midpoint, paired with a red visual cue after playtesting showed the timing change alone wasn't noticeable without a visual signal.
- **Polish pass:** added collect/weave success/fail character animations, a tiered basket result sprite, background music, and pickup/weave sound effects as I worked on other parts of the game. Most of it was added near the end though.